

Zero Speed Sensorless Scheme for PMSM Drives Using Virtual Flux Signal Injection Based Hybrid Active Flux Observer

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Abstract—Sensorless zero-speed control of permanent magnet synchronous motors using high-frequency signal injection introduces acoustic noise and vibration, while conventional model-based observers fail due to unobservability and instability. This article proposes a hybrid active flux observer (HFO) based on virtual flux signal injection method. Compared to conventional sensorless control methods, this method achieves stable zero-speed sensorless control under rated load without high-frequency signal injection, eliminating additional noise and vibration. First, speed observability conditions under virtual flux signal injection are derived using local weak observability theory. Then, the boundary conditions for system stability were derived using the small signal perturbation theory. By injecting virtual flux signal into the current-model flux of the HFO, both local weak observability and stability requirements at zero speed are simultaneously satisfied. Experimental validation on a 5.2 kW interior permanent magnet synchronous motor (IPMSM) drive platform confirms the method's effectiveness.

Index Terms—Model-based method, permanent magnet synchronous motor (PMSM), sensorless control, virtual flux signal injection, zero-speed.

I. INTRODUCTION

AS a core component of high-efficiency electric drive systems, permanent magnet synchronous motors (PMSMs) have become the preferred actuators in industry due to their

Received 5 April 2025; revised 21 July 2025 and 29 October 2025; accepted 8 November 2025. Date of publication 29 December 2025; date of current version 3 April 2026. This work was supported by Hubei Provincial Natural Science Foundation under Grant 2024AFB576. (Corresponding author: Wei Sun)

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Digital Object Identifier 10.1109/TIE.2025.3634413

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high-power density and precise controllability. Traditional vector control schemes relying on mechanical speed sensors are constrained by cost and reliability limitations, thereby driving widespread industrial interest in sensorless schemes [1].

Sensorless control methods mainly fall into two categories: saliency-based observers [2], [3], [4], [5], [6], [7], [8], [9], [10], [11] and model-based observers [13], [14], [15], [16], [17], [18], [19], [20], [21], [22]. Saliency-based observers inject additional high-frequency square wave signal [2], [3], [4], [9], [10] or sinusoidal signal [5], [6], [7], [8], [11] into the motor stator winding, which can address zero/low-speed stable load operation issues. However, traditional high frequency signal injection (HFSI) methods can cause noise and vibration, and may fail due to the low saliency of surface-mounted permanent magnet synchronous motors (SPMSMs) [10]. Due to the system delay caused by using filters to demodulate signals in traditional HFSI methods, the dynamic response of the system will be reduced [9], [11]. Although some modified saliency-based methods can solve these problems, most of them require high computing power platforms [10]. In [12], a method utilizing inverter harmonics to obtain rotor angle is mentioned. This approach does not require injecting additional high-frequency signals but instead demodulates the rotor angle using the inherent high-frequency harmonic signals generated by the inverter. This method enables motor operation under zero-speed heavy-load conditions, but due to the limitations of existing hardware computing capabilities, the carrier frequency of the control system cannot be too high, and low carrier frequencies can also cause noise problems.

Model-based observers rely solely on mathematical model of motor and can adapt to any PMSM type. Nevertheless, these methods are widely recognized as challenging for zero/low-speed heavy-load applications [23], [18]. In [24], [25], it was point out that stator current time-varying constitute a necessary condition for observer system observability at zero speed. In [26], [27], [28], researchers reveal that unstable poles emerge in observer systems during zero/low-speed operation, leading to instability of model-based observers. Additionally, most model-based observers exhibit sensitivity to nonlinear factors of power switching devices and motor parameter variations [29], which also limits their practical application.

TABLE I
HORIZONTAL COMPARISON BETWEEN EXISTING METHODS AND THE PROPOSED METHOD

Methods	Minimum Stable Speed at Rated Load	Maximum Stable Load at Zero Speed	Angle Estimation Accuracy at Zero Speed under Maximum Stable Load	Does the Method Require Filters to Demodulate Signals?	Does the Method Exist High Frequency Noise?	Does the Method Necessitate the Motor's Saliency Ratio?
Pseudorandom HF square-wave injection [9]	0	100%	$\geq 10^\circ$	No	Yes (20% carrier frequency)	Yes
HF square-wave injection with current oversample [10]	0	100%	$\geq 13^\circ$	No	Yes (100% carrier frequency)	No
HF sine-wave injection with time-domain amplitude demodulator [11]	0	100%	$\geq 10^\circ$	No	Yes (6% carrier frequency)	Yes
Decoupled sliding-mode control method [17]	0	100% (without theoretical proof in the article)	$\geq 10^\circ$	No	No	No
Adaptive stator resistance & load compensation [20]	Approaching 0	$\geq 30\%$	$\geq 15^\circ$	No	No	No
The proposed method	0	100%	Approaching 90°	No	No	No

Over past decades, researchers have proposed various improved model-based observers to expand the operational range of model-based observers. In [18], it was suggested that injecting d -axis current pulses in the low-speed region when using flux observers can effectively enhance load capacity. However, this approach essentially remains a high-frequency injection method. In [19], the authors argued that inaccurate model parameters could lead to increased estimation errors in observers, thereby affecting system stability. They improved estimation accuracy by considering magnetic saturation and flux linkage dynamics. In [20], the authors identified that sampling errors and the initial dc bias of integrators could cause instability in active flux observer at low speeds. They proposed further enhancing speed estimation accuracy through compensation of estimated load torque and performing stator resistance identification, thereby significantly improving low-speed performance. While these methods have achieved certain improvements in enhancing the low-speed load performance of motors, stable load operation remains challenging when approaching zero speed due to theoretical unobservability or instability.

In [21], the authors proposed injecting positive and negative virtual flux signals into the output signal of the flux observer (i.e. the input of the phase-locked loop), so that the motor parameter identification model is not affected by position error changes and inverter nonlinearity at different speeds and loads. However, this article does not solve the problem of unobservability and instability of PMSM speed sensorless control system at zero speed. In [13], [22], the application of virtual voltage injection in induction motors is discussed. The authors point out that the instability of the sensorless zero-speed control system in the induction motor arises from the unobservability of the rotor speed and the presence of poles with positive real parts in the system. The authors inject virtual voltage signal into the observer to restore the local weak observability of the induction motor state variables at zero speed, thereby achieving zero speed stable control of the induction motor.

This article proposes a zero-speed sensorless control method for PMSMs based on a hybrid active flux observer (HFO) with virtual flux signal injection. First, the nonlinear local weak observability theory is employed to analyze the observability conditions of PMSMs with and without virtual flux signal injection. Then, the small-signal perturbation method is utilized to derive the boundary conditions for virtual flux signal injection that ensure system stability. By integrating the observability and stability conditions, the range of virtual flux signal injection satisfying both observability and stability at zero speed is determined. The horizontal comparison between the existing method and the proposed method is shown in Table I. Compared with traditional methods, this method is simple to implement and enables zero-speed operation under rated load without injecting additional high-frequency signal into the motor, and it has good robustness of motor parameters. Finally, the effectiveness of the proposed method was verified on a 5.2 kW interior permanent magnet synchronous motor (IPMSM) drive platform.

II. OBSERVABILITY ANALYSIS OF HYBRID ACTIVE FLUX OBSERVER AT ZERO SPEED

A. Mathematical Model of Hybrid Active Flux Observer for PMSM

The HFO is detailed in [14], when the switch in Fig. 1 is moved to the position marked '0', it represents the traditional HFO system block diagram. The proportional integral (PI) regulator can be simplified to proportional regulator, where the proportional coefficient is defined as $K_p = K_{ob}$.

The current model flux in the rotating frame of the HFO can be written as

$$\begin{bmatrix} \hat{\psi}_d^C \\ \hat{\psi}_q^C \end{bmatrix} = \begin{bmatrix} \hat{L}_d & 0 \\ 0 & \hat{L}_q \end{bmatrix} \begin{bmatrix} \hat{i}_d \\ \hat{i}_q \end{bmatrix} + \begin{bmatrix} \hat{\psi}_f \\ 0 \end{bmatrix}. \quad (1)$$

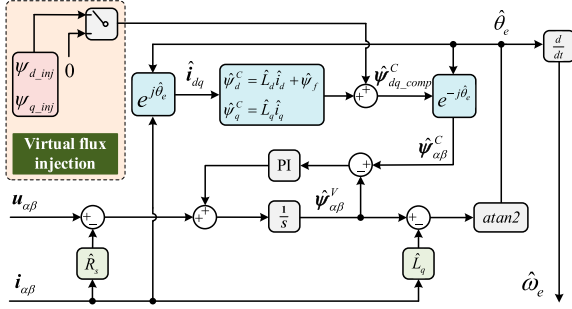


Fig. 1. HFO control block diagram based on virtual flux signal injection.

The voltage model flux in the stationary frame of the HFO can be written as

$$\frac{d}{dt} \begin{bmatrix} \hat{\psi}_\alpha^V \\ \hat{\psi}_\beta^V \end{bmatrix} = \begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} - \hat{R}_s \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + K_{ob} \left(\begin{bmatrix} \hat{\psi}_\alpha^C \\ \hat{\psi}_\beta^C \end{bmatrix} - \begin{bmatrix} \hat{\psi}_\alpha^V \\ \hat{\psi}_\beta^V \end{bmatrix} \right). \quad (2)$$

Transforming (2) into estimated rotating frame yields

$$(p + J\hat{\omega}_e) \begin{bmatrix} \hat{\psi}_d^V \\ \hat{\psi}_q^V \end{bmatrix} = \begin{bmatrix} \hat{u}_d \\ \hat{u}_q \end{bmatrix} - \hat{R}_s \begin{bmatrix} \hat{i}_d \\ \hat{i}_q \end{bmatrix} + K_{ob} \left(\begin{bmatrix} \hat{\psi}_d^C \\ \hat{\psi}_q^C \end{bmatrix} - \begin{bmatrix} \hat{\psi}_d^V \\ \hat{\psi}_q^V \end{bmatrix} \right) \quad (3)$$

where $\hat{R}_s, \hat{L}_d, \hat{L}_q, \hat{\psi}_f$ are the nominal values of stator resistance, d -axis inductance, q -axis inductance, and rotor flux, respectively;

$\hat{\psi}_d^C, \hat{\psi}_q^C$ represents the current model flux in estimated rotating frame;

$\hat{\psi}_\alpha^C, \hat{\psi}_\beta^C$ represents the current model flux in estimated stationary frame;

$\hat{\psi}_\alpha^V, \hat{\psi}_\beta^V$ represents the voltage model flux in estimated stationary frame;

$\psi_{d_inj}, \psi_{q_inj}$ represents the injection value of virtual flux signal;

$\hat{i}_d, \hat{i}_q$ represents the current in estimated rotating frame;

$\hat{u}_d, \hat{u}_q$ represents the voltage in estimated rotating frame;

i_α, i_β represents the current in stationary frame;

u_α, u_β represents the voltage in stationary frame;

$\hat{\omega}_e$ represents the estimated electrical angular velocity;

p represents the differential operator;

$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

In the following text, bold symbols represent the vector form of state variables. State variables with “ $\wedge$ ” represent values in the estimated frame, while state variables without “ $\wedge$ ” represent values in the actual frame.

B. Local Weak Observability Theory for Nonlinear Systems

The theory of local weak observability for nonlinear systems is only briefly explained here, and detailed content such as calculation methods can be found in [30].

Assume a nonlinear dynamical system Σ

$$\Sigma: \begin{cases} \frac{dx}{dt} = f(x, u) \\ y = h(x) \end{cases} \quad (4)$$

where $x \in \Omega \subset \mathbb{R}^n$ is the state vector, $u \in U \subset \mathbb{R}^m$ is the vector of the input values, $y \in \mathbb{R}^r$ is the output vector, and f and h are nonlinear functions. Assuming $x_0 \in \Omega$ is a point in the state space Σ , let $J = \partial L / \partial x_{x=x_0}$, where $L = [\mathcal{L}_f^0 h \quad \mathcal{L}_f h \quad \dots \quad \mathcal{L}_f^{n-1} h]^T$, J is the Jacobian matrix defined as the observability criterion matrix, and $\mathcal{L}_f^k h$ is the k th-order Lie derivative of function h with respect to the vector field f . If matrix J has full rank, then the state variables of system Σ is locally weakly observable at point x_0 .

C. Observability Analysis of Hybrid Active Flux Observer

To avoid excessive length, this section directly incorporates the virtual flux signal injection proposed in this article for analysis (when analyzing results without virtual flux signal injection, simply set the injection signal value to 0). The virtual flux signal is injected into the current model flux (1) in the rotating frame

$$\begin{bmatrix} \hat{\psi}_d^C \\ \hat{\psi}_q^C \end{bmatrix} = \begin{bmatrix} \hat{L}_d & 0 \\ 0 & \hat{L}_q \end{bmatrix} \begin{bmatrix} \hat{i}_d \\ \hat{i}_q \end{bmatrix} + \begin{bmatrix} \hat{\psi}_f \\ 0 \end{bmatrix} + \begin{bmatrix} \psi_{d_inj} \\ \psi_{q_inj} \end{bmatrix}. \quad (5)$$

Then (3) can be rewritten as

$$\begin{cases} \frac{d\hat{\psi}_d^V}{dt} = \hat{u}_d - \hat{R}_s \hat{i}_d + K_{ob} \hat{L}_d \hat{i}_d + K_{ob} (\hat{\psi}_f + \psi_{d_inj}) - K_{ob} \hat{\psi}_d^V \\ \quad + \hat{\omega}_e \hat{\psi}_q^V \\ \frac{d\hat{\psi}_q^V}{dt} = \hat{u}_q - \hat{R}_s \hat{i}_q + K_{ob} \hat{L}_q \hat{i}_q + K_{ob} \psi_{q_inj} - K_{ob} \hat{\psi}_q^V - \hat{\omega}_e \hat{\psi}_d^V \end{cases} \quad (6)$$

The voltage model flux in the observer's rotating frame is selected as the state variable. When analyzing system observability, it is assumed that the electrical parameters of the motor are known and constant, the load torque, rotor and load moment of inertia are unknown, and the rotor speed is constant or slowly changing. Then

$$\Sigma: \begin{cases} \hat{x} = \begin{bmatrix} \hat{\psi}_d^V & \hat{\psi}_q^V & \hat{\omega}_e \end{bmatrix}^T \\ \hat{u} = \begin{bmatrix} \hat{u}_d & \hat{u}_q & \hat{i}_d & \hat{i}_q \end{bmatrix}^T \\ \frac{d\hat{x}}{dt} = f(\hat{x}, \hat{u}) = \begin{bmatrix} \frac{d\hat{\psi}_d^V}{dt} & \frac{d\hat{\psi}_q^V}{dt} & \frac{d\hat{\omega}_e}{dt} \end{bmatrix}^T \\ y = h(\hat{x}) = [h_1 \quad h_2]^T = \begin{bmatrix} \hat{\psi}_d^V & \hat{\psi}_q^V \end{bmatrix}^T \end{cases} \quad (7)$$

where

$$f(\hat{x}, \hat{u}) = \begin{bmatrix} \hat{u}_d - \hat{R}_s \hat{i}_d + K_{ob} \hat{L}_d \hat{i}_d + K_{ob} (\hat{\psi}_f + \psi_{d_inj}) - K_{ob} \hat{\psi}_d^V + \hat{\omega}_e \hat{\psi}_q^V \\ \hat{u}_q - \hat{R}_s \hat{i}_q + K_{ob} \hat{L}_q \hat{i}_q + K_{ob} \psi_{q_inj} - K_{ob} \hat{\psi}_q^V - \hat{\omega}_e \hat{\psi}_d^V \\ 0 \end{bmatrix}. \quad (8)$$

The state space order of the system is 3, and the observability criterion matrix J can be obtained from (8) as follows:

$$J = \frac{\partial L}{\partial \hat{x}} = \begin{bmatrix} \frac{\partial \mathcal{L}_f^0 h_1}{\partial \hat{\psi}_d^V} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \hat{\psi}_q^V} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \hat{\omega}_e} \\ \frac{\partial \mathcal{L}_f^1 h_2}{\partial \hat{\psi}_d^V} & \frac{\partial \mathcal{L}_f^1 h_2}{\partial \hat{\psi}_q^V} & \frac{\partial \mathcal{L}_f^1 h_2}{\partial \hat{\omega}_e} \\ \vdots & \vdots & \vdots \\ \frac{\partial \mathcal{L}_f^2 h_2}{\partial \hat{\psi}_d^V} & \frac{\partial \mathcal{L}_f^2 h_2}{\partial \hat{\psi}_q^V} & \frac{\partial \mathcal{L}_f^2 h_2}{\partial \hat{\omega}_e} \end{bmatrix}. \quad (9)$$

To make $\text{rank}\{J\} = n = 3$, (9) requires at least one 3×3 submatrix with full rank, with a total of 20 possible outcomes. Rows 1, 2, and 5 are taken as the validation submatrix J_1

$$J_1 = \begin{bmatrix} \frac{\partial \mathcal{L}_f^0 h_1}{\partial \psi_d^V} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \psi_q^V} & \frac{\partial \mathcal{L}_f^0 h_1}{\partial \omega_e} \\ \frac{\partial \mathcal{L}_f^0 h_2}{\partial \psi_d^V} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial \psi_q^V} & \frac{\partial \mathcal{L}_f^0 h_2}{\partial \omega_e} \\ \frac{\partial \mathcal{L}_f^2 h_1}{\partial \psi_d^V} & \frac{\partial \mathcal{L}_f^2 h_1}{\partial \psi_q^V} & \frac{\partial \mathcal{L}_f^2 h_1}{\partial \omega_e} \end{bmatrix}. \quad (10)$$

According to (10), to make J_1 full rank, it only needs that the determinant of J_1 is not equal to 0.

$$\det(J_1) = 2\hat{\omega}_e(\hat{L}_d \hat{i}_d + \hat{\psi}_f - \hat{\psi}_d^V) + 2K_{ob}(\hat{L}_q \hat{i}_q - \hat{\psi}_q^V) + \hat{L}_q \frac{d\hat{i}_q}{dt} + K_{ob}\psi_{q_inj}. \quad (11)$$

When the speed is 0, since $\hat{L}_q \hat{i}_q$ it is the q -axis current model flux, when the hybrid flux observer converges, the current model flux is equal to the voltage model flux. and if consider $d\hat{i}_q/dt = 0$, then

$$\det(J_1) = K_{ob}\psi_{q_inj}. \quad (12)$$

From (12), it can be seen that when there is no virtual flux injection, $\det(J_1) = 0$, this situation does not meet the local weak observability condition. Since $K_{ob} > 0$, only $\psi_{q_inj} \neq 0$ is needed to make HFO locally weak observability at zero speed. The specific range of its values needs further analysis of its stability conditions, the problem of sensorless stable control of PMSM at zero speed can only be solved when both observability and stability conditions are met.

III. STABILITY ANALYSIS AND PARAMETERS ROBUSTNESS ANALYSIS OF ACTIVE FLUX OBSERVER

A. Stability Analysis Based on Small Signal Perturbation Method

The angle estimation error $\tilde{\theta}_e = \hat{\theta}_e - \theta_e$ between the estimated angle $\hat{\theta}_e$ of HFO and the real angle θ_e , which is feedback into the observer system through the current model flux and thus affects the stability of the system. Transform the observer equation from stationary frame to rotating frame and use the small signal perturbation method to derive the linearized small signal model of the observer system. Assuming that, PMSM operates at a static operating point, in rotating frame, the variables such as voltage, current, and flux in Fig. 1 will be shown in the form of direct flow and considered as steady-state operating points. Based on the steady-state operating point, small signal disturbances are added to each variable x of the system.

$$x = x_0 + \Delta x. \quad (13)$$

The subscript “0” represents the value of the steady-state operating point. After removing the dc steady-state term and high-order small signal term generated during the calculation process, the small signal model of the system can be obtained by linearization.

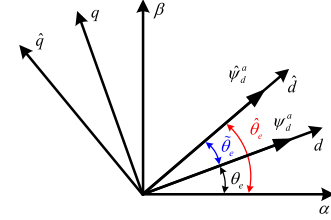


Fig. 2. Schematic diagram of the difference between the estimated frame and the actual frame.

Add virtual flux signal injection to the current model flux and transform it into the real rotating frame yields

$$\psi_{dq_comp}^C = P(\theta_e)P^{-1}(\hat{\theta}_e) \left(\hat{L}_{dq}P(\hat{\theta}_e)P^{-1}(\theta_e)i_{dq} + \begin{bmatrix} \hat{\psi}_f \\ 0 \end{bmatrix} + \begin{bmatrix} \psi_{d_inj} \\ \psi_{q_inj} \end{bmatrix} \right). \quad (14)$$

Transforming (1) into real rotating frame yields

$$(p + J\omega_e)\psi_{dq}^V = u_{dq} - \hat{R}_s i_{dq} + K_{ob}(\psi_{dq_comp}^C - \psi_{dq}^V). \quad (15)$$

Take the small signal model for (15)

$$(p + J\omega_e)\Delta\psi_{dq}^V = \Delta u_{dq} - \hat{R}_s \Delta i_{dq} + K_{ob}(\Delta\psi_{dq_comp}^C - \Delta\psi_{dq}^V) \quad (16)$$

where $\psi_{dq_comp}^C$ is the current model flux vector injected with virtual flux. Symbols preceded by Δ denotes the small signal model

$$P(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}, P^{-1}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

From Fig. 2, it can be seen that the active flux vector $\vec{\psi}_d^a = \vec{\psi}_s - \hat{L}_q \vec{i}_s = (\psi_d^V - \hat{L}_q i_d) + j(\psi_q^V - \hat{L}_q i_q)$. The angle estimation error can be obtained as

$$\tilde{\theta}_e = \hat{\theta}_e - \theta_e = \arctan \left(\frac{\psi_q^V - \hat{L}_q i_q}{\psi_d^V - \hat{L}_q i_d} \right). \quad (17)$$

Assuming the estimated angle is relatively accurate, (17) can be approximated as:

$$\tilde{\theta}_e = \hat{\theta}_e - \theta_e \approx \frac{\psi_q^V - \hat{L}_q i_q}{\psi_d^V - \hat{L}_q i_d}. \quad (18)$$

Take the small signal model for (18)

$$\Delta\tilde{\theta}_e = \frac{\Delta\psi_q^V - \hat{L}_q \Delta i_q}{\psi_d^V - \hat{L}_q i_{d0}} = \frac{\Delta\psi_q^V - \hat{L}_q \Delta i_q}{(\hat{L}_d - \hat{L}_q)i_{d0} + \hat{\psi}_f}. \quad (19)$$

Combining (19), the small signal model of (14) can be obtained as

$$\begin{bmatrix} \Delta\psi_{dq_comp}^C \\ \Delta\psi_{dq_comp}^C \end{bmatrix} = \begin{bmatrix} \left[(\hat{L}_d - \hat{L}_q)i_{q0} - \psi_{q_inj0} \right] \frac{\Delta\psi_q^V - \hat{L}_q \Delta i_q}{(\hat{L}_d - \hat{L}_q)i_{d0} + \hat{\psi}_f} + \hat{L}_d \Delta i_d + \Delta\psi_{d_inj} \\ \left[\Delta\psi_q^V + \psi_{d_inj0} \right] \frac{\Delta\psi_q^V - \hat{L}_q \Delta i_q}{(\hat{L}_d - \hat{L}_q)i_{d0} + \hat{\psi}_f} + \Delta\psi_{q_inj} \end{bmatrix}. \quad (20)$$

Substituting (20) into (16) yields the small signal model of HFO in rotating frame

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} \Delta\psi_d^V \\ \Delta\psi_q^V \end{bmatrix} &= \begin{bmatrix} -K_{ob} & K_1 + \omega_e \\ -\omega_e & K_2 \end{bmatrix} \begin{bmatrix} \Delta\psi_d^V \\ \Delta\psi_q^V \end{bmatrix} \\ &+ \begin{bmatrix} K_{ob}\hat{L}_d - \hat{R}_s & -\hat{L}_q K_1 \\ 0 & -\hat{R}_s - \hat{L}_q K_2 \end{bmatrix} \begin{bmatrix} \Delta i_d \\ \Delta i_q \end{bmatrix} \\ &+ K_{ob} \begin{bmatrix} \Delta\psi_{d_inj} \\ \Delta\psi_{q_inj} \end{bmatrix} + \begin{bmatrix} \Delta u_d \\ \Delta u_q \end{bmatrix} \end{aligned} \quad (21)$$

where

$$\begin{cases} \hat{\psi}_d^a = (\hat{L}_d - \hat{L}_q)i_{d0} + \hat{\psi}_f \\ K_1 = \frac{K_{ob}[(\hat{L}_d - \hat{L}_q)i_q - \psi_{q_inj0}]}{\hat{\psi}_d^a} \\ K_2 = \frac{K_{ob}\psi_{d_inj0}}{\hat{\psi}_d^a} \end{cases} \quad (22)$$

Since the virtual flux signal is equivalent to adding a dc bias to the current model flux of HFO, then in (21), $\Delta\psi_{d_inj} = \Delta\psi_{q_inj} = 0$, and let $\psi_{dq_inj0} = \psi_{dq_inj}$, $i_{dq0} = i_{dq}$. So (21) is the standard state equation form, with its coefficient matrix A is as follows:

$$A = \begin{bmatrix} -K_{ob} & \frac{K_{ob}[(\hat{L}_d - \hat{L}_q)i_q - \psi_{q_inj}]}{\hat{\psi}_d^a} + \omega_e \\ -\omega_e & \frac{K_{ob}\psi_{d_inj}}{\hat{\psi}_d^a} \end{bmatrix}. \quad (23)$$

The characteristic equation of the system can be obtained through the coefficient matrix (23) as follows:

$$\begin{aligned} |sI - A| &= \begin{vmatrix} s + K_{ob} & -\frac{K_{ob}[(\hat{L}_d - \hat{L}_q)i_q - \psi_{q_inj}]}{\hat{\psi}_d^a} - \omega_e \\ \omega_e & s - \frac{K_{ob}\psi_{d_inj}}{\hat{\psi}_d^a} \end{vmatrix} \\ &= s^2 + \left(K_{ob} - \frac{K_{ob}\psi_{d_inj}}{\hat{\psi}_d^a} \right) s \\ &\quad - \frac{K_{ob}^2\psi_{d_inj}}{\hat{\psi}_d^a} + \frac{\omega_e K_{ob}[(\hat{L}_d - \hat{L}_q)i_q - \psi_{q_inj}]}{\hat{\psi}_d^a} + \omega_e^2 \\ &= 0. \end{aligned} \quad (24)$$

It is important to note that the steady-state equations neglected during small-signal perturbation analysis correspond to different steady-state operating points of the motor. So, the stability analysis of the motor system focuses more on the stable regions of the system.

When the real parts of both roots of the characteristic equation are less than 0, the system is stable. By introducing the virtual flux injection, the stable regions of the system can be altered to achieve global stability. At this point, the corresponding range of values for the virtual flux signal injection can be determined as follows:

$$\begin{cases} \psi_{d_inj} < 0 \\ \Delta\hat{L}_q - 2\sqrt{-\psi_{d_inj}\hat{\psi}_d^a} < \psi_{q_inj} < \Delta\hat{L}_q + 2\sqrt{-\psi_{d_inj}\hat{\psi}_d^a} \end{cases} \quad (25)$$

Since it is necessary to ensure stability from no-load to full load at zero speed, (25) needs to take the intersection of the boundary conditions of virtual flux signal injection under electric rated load, power generation rated load, and no-load.

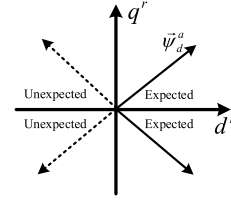


Fig. 3. Schematic diagram of active flux vector in different quadrants.

Then by combining the observability conditions and (25), the range of values for injecting virtual flux signal that satisfy both observability and stability in PMSMs can be obtained as follows:

$$\begin{cases} \psi_{d_inj} < 0 \\ \psi_{q_inj} \neq 0 \\ -\sqrt{2}\Delta\hat{L}I_N - 2\sqrt{-\psi_{d_inj}\hat{\psi}_d^a} < \psi_{q_inj} < \sqrt{2}\Delta\hat{L}I_N \\ +2\sqrt{-\psi_{d_inj}\hat{\psi}_d^a} \end{cases} \quad (26)$$

where I_N is the rated current of the motor, $\Delta\hat{L} = \hat{L}_d - \hat{L}_q$.

B. Design Methodology for Virtual Flux Signal Injection

From (26), it can be seen that the d -axis virtual flux injection can only choose to inject negative values, while the q -axis virtual flux injection can ensure observability and stability conditions regardless of whether positive or negative values are injected. So, the injection form of the virtual flux signal can be selected as follows:

$$\begin{cases} \psi_{d_inj} = K_{inj d}\hat{\psi}_f \\ \psi_{q_inj} = K_{inj q}\hat{\psi}_f \end{cases} \quad (27)$$

where $K_{inj d} < 0$, which means injecting a negative virtual flux signal into the d -axis flux, and the sign of the injection value of the q -axis virtual flux depends on the sign of $K_{inj q}$.

Taking the actual rotor rotation coordinate system $d^r - q^r$ axis as a reference, the active flux vector injected with virtual flux signal can appear in any quadrant by changing the injection value. The angle θ_{inj} between the active flux vector injected with virtual flux signal and the d^r axis can be expressed as

$$\begin{aligned} \theta_{inj} &= \arctan \left(\frac{\psi_{q_inj}}{(\hat{L}_d - \hat{L}_q)i_d + \hat{\psi}_f + \psi_{d_inj}} \right) \\ &= \arctan \left(\frac{K_{inj q}\hat{\psi}_f}{(\hat{L}_d - \hat{L}_q)i_d + \hat{\psi}_f + K_{inj d}\hat{\psi}_f} \right). \end{aligned} \quad (28)$$

However, if the active flux vector is in quadrant 2 or 3 as shown in Fig. 3, the control law of motor current on torque is opposite. At this time, the positive q -axis current represents the power generation load, and the negative q -axis current represents the electric load. Generally, this phenomenon is not desirable.

So, there is $(\hat{L}_d - \hat{L}_q)i_d + \hat{\psi}_f + K_{inj d}\hat{\psi}_f > 0$. Therefore, the range of $K_{inj d}$ is shown as follows:

$$-\frac{(\hat{L}_d - \hat{L}_q)i_d + \hat{\psi}_f}{\hat{\psi}_f} < K_{inj d} < 0. \quad (29)$$

TABLE II
IPMSM PARAMETERS

Parameter	Value	Parameter	Value
Rated power	5.2 kW	Number of pole pairs	3
Rated speed	2000 r/min	Stator resistance	1.39 Ω
Rated torque	25 N·m	d -axis inductance	8.85 mH
Rated voltage	380 V	q -axis inductance	17.7 mH
Rated current	10.6 A	Rotor flux	0.3976 Wb

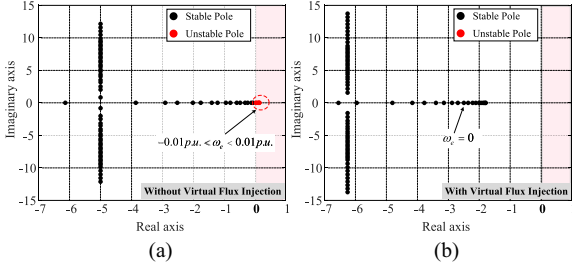


Fig. 4. Pole distribution diagram of HFO. (a) Without virtual flux signal injection. (b) With virtual flux signal injection.

It's recommended that $K_{inj} > 0.1$. To avoid adding too many parameters that need to be adjusted in the observer, the form of virtual flux signal injection in the experiment is designed as (30)

$$\begin{cases} \psi_{d_inj} = -K_{inj}\hat{\psi}_f \\ \psi_{q_inj} = K_{inj}\hat{\psi}_f \\ 0.1 \leq K_{inj} < [(\hat{L}_d - \hat{L}_q)i_d + \hat{\psi}_f]/\hat{\psi}_f \end{cases} \quad (30)$$

In addition, the d -axis current command is adaptively given based on the load size at low speed, as the speed increases, the d -axis current decreases linearly to 0. The specific implementation method can be found in [32].

Based on characteristic equations (24) and (30), the pole distribution of the HFO system can be obtained under two different conditions: without virtual flux signal injection and with virtual flux signal injection. The motor operates under rated load condition, with its rotational speed varying within the range of -0.02 – 0.02 p.u., the value of K_{inj} is 0.1. The motor parameters are provided in Table II.

As shown in Fig. 4(a), the conventional HFO exhibits unstable poles at and near zero speed, leading to system divergence and making it challenging to achieve stable operation with load at zero speed. In contrast, as shown in Fig. 4(b), after incorporating virtual flux injection, the system's unstable poles disappear, allowing the system to maintain stability at and near zero speed.

C. Robustness Analysis Considering Motor Parameter Errors

When considering motor parameter errors from the stability perspective, (24) can be written as:

$$|sI - A|$$

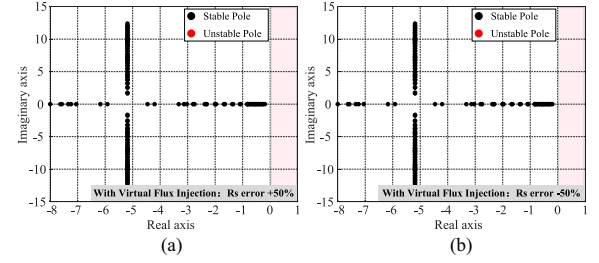


Fig. 5. Pole distribution diagram of R_s error $\pm 50\%$ under injection of virtual flux signal. (a) R_s error $+50\%$. (b) R_s error -50% .

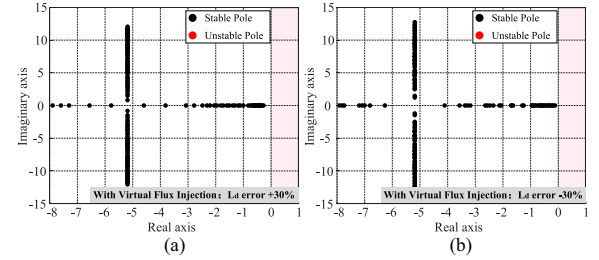


Fig. 6. Pole distribution diagram of L_d error $\pm 30\%$ under injection of virtual flux signal. (a) L_d error $+30\%$. (b) L_d error -30% .

$$= \begin{vmatrix} s + K_{ob} & -\frac{K_{ob}[(L_d + L_{derr} - L_q - L_{qerr})i_d - \psi_{q_inj}]}{(L_d + L_{derr} - L_q - L_{qerr})i_d + \psi_f + \psi_{ferr}} - \omega_e \\ \omega_e & s - \frac{K_{ob}\psi_{d_inj}}{(L_d + L_{derr} - L_q - L_{qerr})i_d + \psi_f + \psi_{ferr}} \end{vmatrix} = 0. \quad (31)$$

The range of virtual flux injection considering parameter errors can be obtained as follows:

$$\begin{cases} \psi_{d_inj} < 0, \psi_{q_inj} \neq 0 \\ -\sqrt{2}\Delta L_{err}I_N - 2\sqrt{-\psi_{d_inj}\hat{\psi}_{d_err}^a} < \psi_{q_inj} < \sqrt{2}\Delta L_{err}I_N \\ + 2\sqrt{-\psi_{d_inj}\hat{\psi}_{d_err}^a} \\ \Delta L_{err} = (L_d + L_{qerr}) - (L_q + L_{qerr}) \\ \hat{\psi}_{d_err}^a = [(L_d + L_{qerr}) - (L_q + L_{qerr})]i_d + \psi_f + \psi_{ferr} \end{cases} \quad (32)$$

where L_d, L_q, ψ_f represents the actual motor parameters, and $L_{derr}, L_{qerr}, \psi_{ferr}$ represents the error of motor parameters.

According to (32), the parameters that affect system stability are mainly inductance and rotor flux parameters. And magnetic saturation is easily generated in the q -axis, which causes the q -axis inductance to decrease with increasing load, but the variation is generally within 30% [31].

Figs. 5–8 respectively shows the pole distribution diagrams of HFO with stator resistance error $\pm 50\%$, dq -axis inductance, and rotor flux error $\pm 30\%$ under the condition of injecting virtual flux signal.

From the above pole distribution diagram, it can be seen that under the injection of virtual flux signal, the stator resistance error is within $\pm 50\%$, and the dq -axis inductance and magnetic flux error are within $\pm 30\%$, the system can still maintain stability.

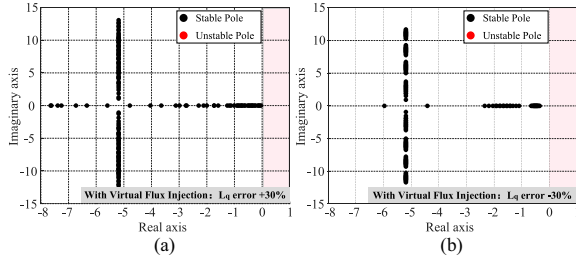


Fig. 7. Pole distribution diagram of L_q error $\pm 30\%$ under injection of virtual flux signal. (a) L_q error $+30\%$. (b) L_q error -30% .

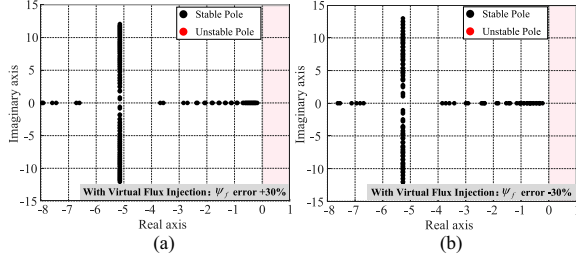


Fig. 8. Pole distribution diagram of ψ_f error $\pm 30\%$ under injection of virtual flux signal. (a) ψ_f error $+30\%$. (b) ψ_f error -30% .

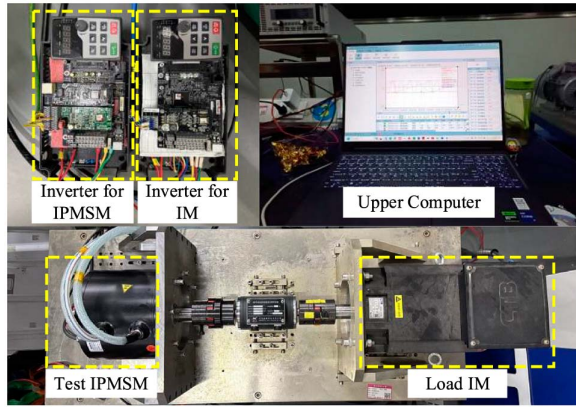


Fig. 9. 5.2 kW IPMSM drive platform.

IV. EXPERIMENTAL RESULT

A. Experimental Platform

Fig. 9 shows the 5.2 kW IPMSM drive platform, which consists of an interior PMSM for testing the proposed algorithm and an induction motor that provides load torque. An encoder was installed on the IPMSM rotor as a real angle comparison. In the following experiment, θ represents the error between the angle estimated by the observer and the angle measured by the encoder. The parameters of the IPMSM are listed in Table II. The control chip is TMS320F28075, with a pulse width modulation (PWM) carrier frequency of 8 kHz and a dead time of 2.4 μ s.

B. Experimental Results and Description

Fig. 10 shows the comparison of experimental results under half load without and with virtual flux signal injection. The motor speed is reduced from 400 r/min to 0. As shown in

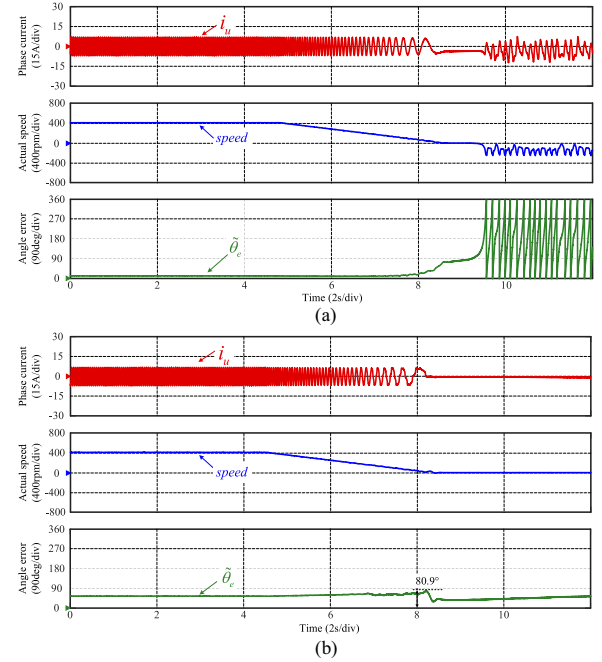


Fig. 10. Comparison of experimental results without/with virtual flux signal injection under 50% load. (a) Without virtual flux signal injection. (b) With virtual flux signal injection.

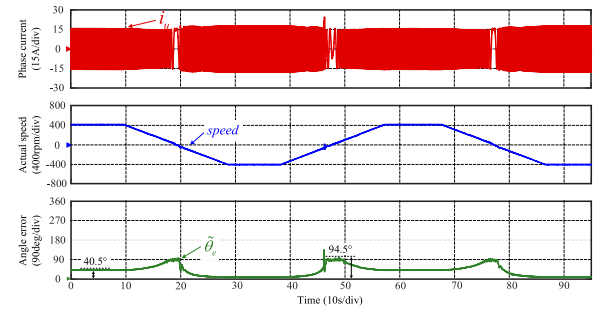


Fig. 11. Experimental results of rated load forward and reverse rotation with virtual flux signal injection.

Fig. 10(a), when the motor speed is reduced to near 0, the estimated angle gradually diverges and is dragged away by the torque applied by the load induction motor (IM). This is due to the unobservability and instability of the HFO at zero speed. From Fig. 10(b), it can be seen that the motor can maintain stability even when decelerating to zero speed with virtual flux signal injection.

Fig. 11 presents the forward and reverse rotation experiment under rated load with virtual flux signal injection. The IPMSM operates under rated load, with speed alternating between $+400$ and -400 r/min. It can be observed that the motor remains stable throughout the entire process.

Fig. 12 illustrates the forward startup experiment at zero speed under full load with virtual flux signal injection. Initially, the IPMSM is controlled to operate at zero speed, and then the load IM applies the rated torque. Subsequently, the IPMSM accelerates forward to $+400$ r/min before decelerating back to zero speed. It can be observed that the estimated angle does not diverge when the motor decelerates to zero speed. Fig. 13

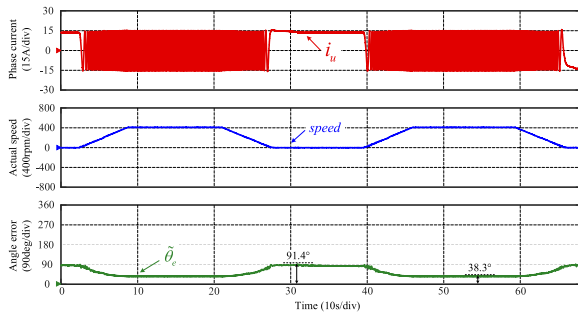


Fig. 12. Experimental results of zero speed forward starting under rated load with virtual flux signal injection.

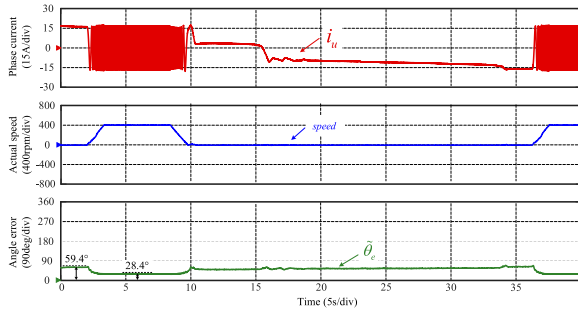


Fig. 13. Experimental results of zero speed rated load starting and hovering with a dead time of 4.8 μ s.

shows the experimental result where all other conditions are the same and only the dead time is increased to 4.8 μ s. By using this method to increase the nonlinear effect of the system, it can be seen that compared to Fig. 12, the stator current is relatively larger. It can be seen that nonlinear effects will to some extent affect the estimation accuracy, but the system is still stable.

Fig. 14 demonstrates the forward and reverse startup experiment at zero speed under rated load with virtual flux signal injection. Initially, the IPMSM is controlled at zero speed under rated torque, then accelerates forward to +400 r/min, decelerates to zero speed and remains there for 23 s before reversing to -400 r/min. Fig. 14(b) presents a zoomed-in view of part A in Fig. 14(a), while Fig. 14(c) shows a zoomed-in view of part B in Fig. 14(a). The experimental results indicate that the proposed method can achieve stable forward and reverse rotation and zero speed starting under rated load.

From Fig. 14 it can be observed that under rated load conditions, as the speed decreases, the angle error gradually increases to around 90°, which is somewhat similar to the effect of IF control. However, this method is fundamentally different from current frequency (IF) control. First, IF control is a current closed-loop and speed open-loop control method. Many commonly used SVC methods in the industry first use IF control mode to drag the motor to a certain speed, so that the observability and stability conditions of the observer are met and then switch to the observer algorithm operation. To achieve starting with different loads during IF control, the rated current is usually directly used as the current command. However, this is actually a switch between two modes, and the observer does not participate in control at zero/low speed. The method of injecting

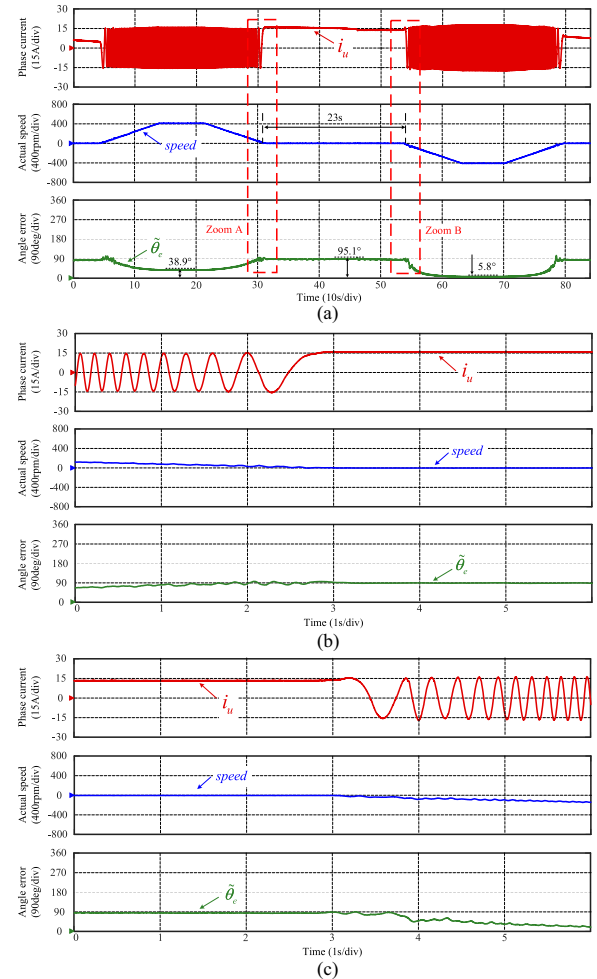


Fig. 14. Experimental results of forward and reverse startup under rated load with virtual flux signal injection. (a) Overall waveform. (b) Zoom A. (c) Zoom B.

virtual flux signal proposed in this article is an observer zero speed closed-loop method, in which the observer participates in the entire control process, which is fundamentally different from IF control.

Fig. 15 demonstrates the sudden loading and unloading experiment at zero speed with virtual flux signal injection. The motor operates at zero speed, suddenly apply the rated load, maintain it for about 5 s, and then suddenly remove the load. Experimental results show that even with sudden loading and unloading at zero speed, the proposed algorithm can still ensure that the estimated angle does not diverge.

Fig. 16 illustrates the variable load experiment with virtual flux signal injection at zero speed. The motor operates at zero speed while the load torque gradually transitions from positive rated torque to negative rated torque. From the experimental results, it can be seen that regardless of the load, the observer never diverges. The experimental results show that the proposed method has good zero speed performance and can ensure the stability at zero speed under load.

Fig. 17 shows the experimental result of different K_{inj} . The motor speed is set to 0 and the load to the rated load. K_{inj} starts

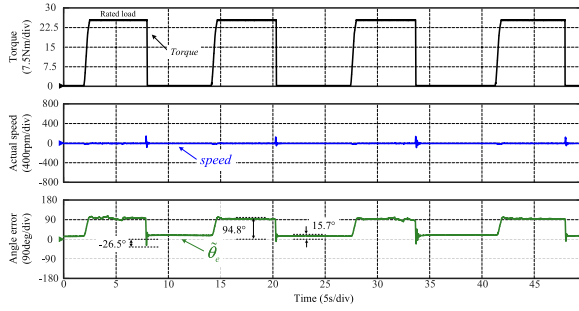


Fig. 15. Experimental results of zero speed sudden loading and unloading with virtual flux signal injection.

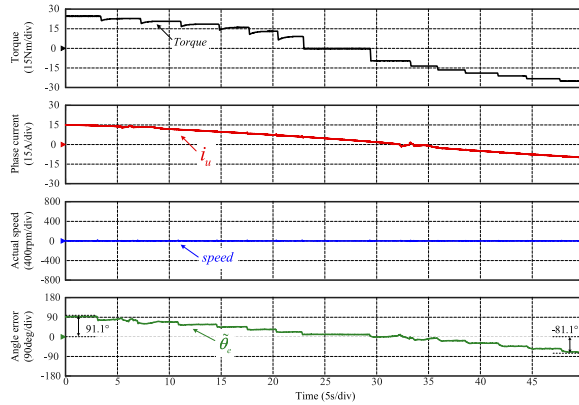


Fig. 16. Experimental results of torque change from positive rated torque to negative rated torque at zero speed with virtual flux signal injection.

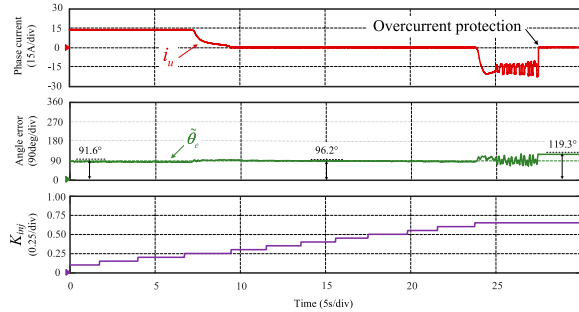


Fig. 17. Experimental result of different K_{inj} with virtual flux signal injection under zero speed rated load condition.

from 0.1 and increases by 0.05 each time. It can be observed that when $K_{inj} = 0.65$, the system begins to diverge until overcurrent protection is triggered and the machine stops. The result indicate that the selection method of injection coefficient can ensure the stability of the system.

Figs. 18–21 respectively shows the motor parameters robustness experiment with virtual flux signal injection. The motor first operates with the correct parameters under rated load, then decelerates to zero speed for about 7.5 s, and then changes the value of the parameters at the dotted line [Increasing the stator resistance to 1.5 times the original value at the dotted line of Fig. 18(a) and reducing it to half the original value at the dotted

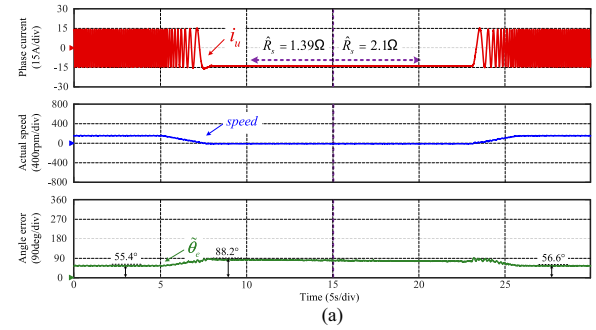


Fig. 18. Experimental results on the robustness of stator resistance parameters with virtual flux signal injection. (a) R_s error +50%. (b) R_s error -50%.

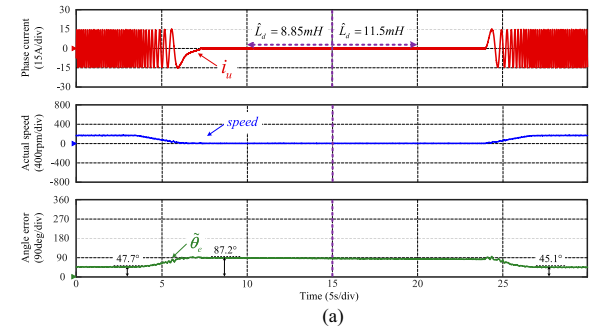
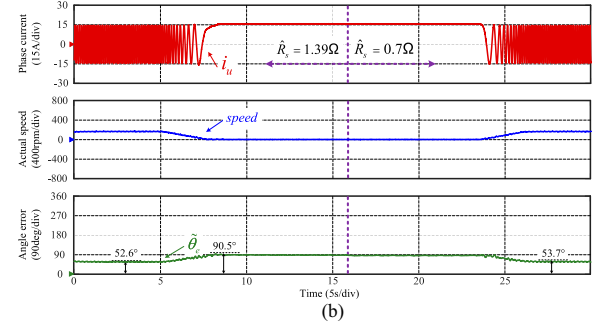
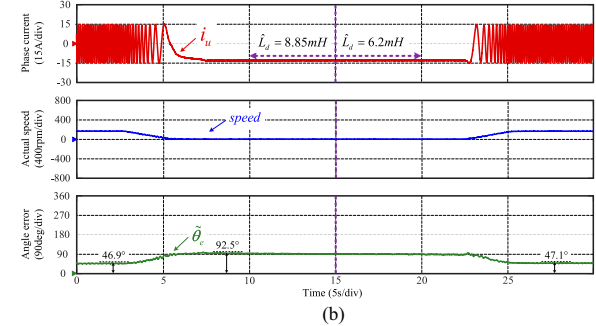


Fig. 19. Experimental results on the robustness of d -axis inductance parameter with virtual flux signal injection. (a) L_d error +30%. (b) L_d error -30%.



line of Fig. 18(b). Increasing the d -axis inductance to 1.3 times the original value at the dotted line of Fig. 19(a) and reducing it to 0.7 times the original value at the dotted line of Fig. 19(b). Increasing the q -axis inductance to 1.3 times the original value at the dotted line of Fig. 20(a) and reducing it to 0.7 times the original value at the dotted line of Fig. 20(b). Increasing the rotor flux to 1.3 times the original value at the dotted line of

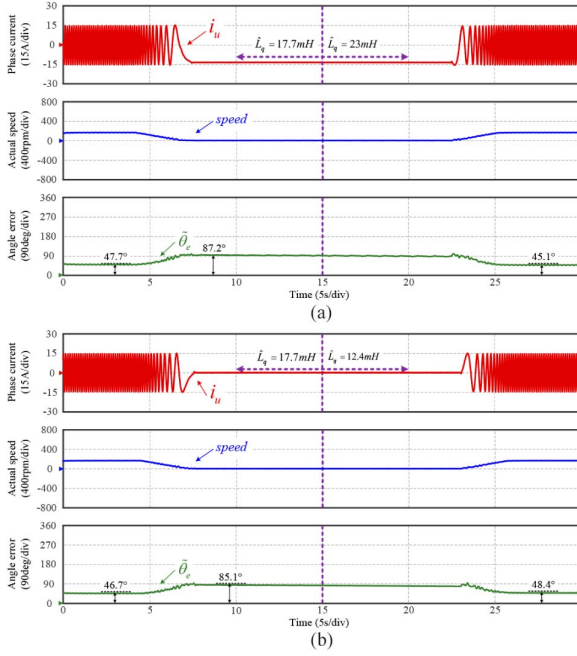


Fig. 20. Experimental results on the robustness of q -axis inductance parameter with virtual flux signal injection. (a) L_q error +30%. (b) L_q error -30%.

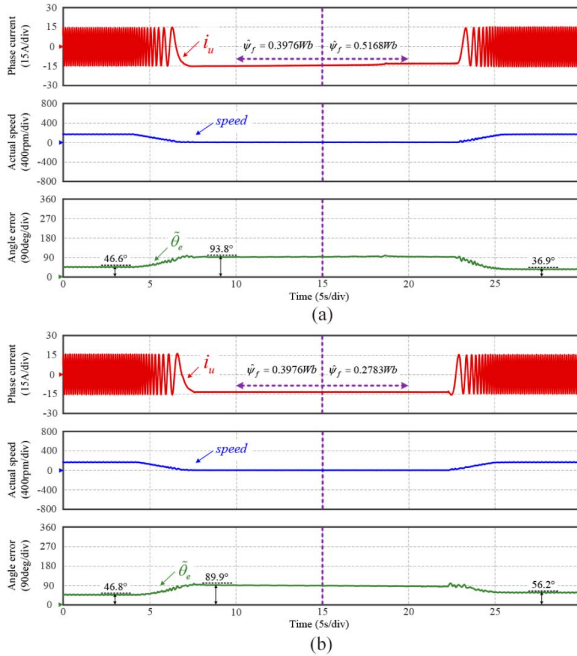


Fig. 21. Experimental results on the robustness of rotor flux parameter with virtual flux signal injection. (a) ψ_f error +30%. (b) ψ_f error -30%.

Fig. 21(a) and reducing it to 0.7 times the original value at the dotted line of Fig. 21(b)]. After changing the parameters, maintain it at zero speed for about 7.5 s before starting the IPMSM. From the experimental results, it can be seen that stator resistance error of $\pm 50\%$, dq -axis inductance and rotor flux error of $\pm 30\%$ has no effect on the stability of the system, the proposed method has good parameters robustness.

V. CONCLUSION

This article proposes an PMSM zero speed sensorless control method based on HFO with virtual flux signal injection. The zero speed observability conditions are analyzed through non-linear local weak observability theory for both the conventional HFO and its improved counterpart employing virtual flux signal injection. On the basis of observability conditions, the boundary conditions for injecting virtual flux signal that can simultaneously satisfy observability and stability at zero speed were analyzed using the small signal perturbation method. According to the relationship between the virtual flux signal and the angle orientation, the boundary of the injection coefficient was analyzed. The experimental results demonstrate that compared with traditional model-based observer methods, this method can achieve stable forward and reverse rotation under load, zero speed loaded starting, zero speed loaded hovering, and zero speed sudden loading and sudden unloading without injecting additional high-frequency signal into the motor stator winding. In addition, this method has good robustness to motor parameters, especially the robustness of stator resistance.

APPENDIX

A. Summary of the Proposed Method

This article proposes a method of injecting virtual flux signal into HFO, which solves the problem of zero speed unobservability and instability in PMSMs. The derivation process relies on certain linearization processing but does not affect the final analysis results and experimental results. Compared to traditional model-based observer methods, the proposed method does not require additional hardware circuits. By injecting virtual flux signal, it can effectively improve the observability and stability of traditional HFO, and broaden the speed control range of model-based sensorless control. In practical operation, it is necessary to select the value of the injected virtual flux signal based on the boundary conditions shown in (30), and according to the method in [32], the d -axis current command is adaptively given based on the load size at low speed.

B. Theoretical Basis for the Proposed Method's Stability Despite Near 90° Angle Errors at Zero Speed

Assuming there is a fixed point $\bar{x}$ in the nonlinear system (33), the behavior in the neighborhood of $\bar{x}$ can be studied by studying the first-order small signal model (34) (linear system) of (33)

$$\dot{x} = f(x), \quad x \in \mathbb{R}^n \quad (33)$$

$$\dot{\xi} = Df(\bar{x})\xi, \quad \xi \in \mathbb{R}^n. \quad (34)$$

There is a conclusion [33]: In the case where f is sufficiently smooth, if and only if all eigenvalues of matrix $Df(\bar{x})$ have negative real parts, $\dot{x} = f(x)$ is exponentially stable at $\bar{x}$.

This conclusion can be proven by the three theorems mentioned in [33], [34].

Theorem 1: If $Df(\bar{x})$ has no eigenvalues with zero real part, $\bar{x}$ is called a hyperbolic or nondegenerate fixed point and the

asymptotic behavior of solutions near it (and hence its stability type) is determined by the linearization. If any one of the eigenvalues has zero real part, then stability cannot be determined by linearization.

Theorem 2: Suppose that $\dot{x} = f(x)$ has a hyperbolic fixed point $\bar{x}$. Then there exist local stable and unstable manifolds $W_{loc}^s(\bar{x}), W_{loc}^u(\bar{x})$, of the same dimensions n_s, n_u as those of the eigenspaces E^s, E^u of the linearized system $\dot{\xi} = Df(\bar{x})\xi$, and tangent to E^s, E^u at $\bar{x}$. $W_{loc}^s(\bar{x}), W_{loc}^u(\bar{x})$ are as smooth as the function f .

Theorem 3: Nonlinear system can be decomposed into a linear part G and a nonlinear part Q , i.e. $S(G, Q)$. For the Q -part, it can be replaced with a set Δ of linear maps which capture its behavior. Then $S(G, Q)$ can be replaced with $S(G, \Delta)$, if the input-output behaviors associated with all the mappings $S(G, \Delta)$ satisfy a given property, then so must any input-output behavior of $S(G, Q)$.

When use the small signal perturbation method to analyze the stability of the observer system, it is essentially an example based on the above conclusion. Factors such as parameter errors, load disturbances, and modeling inaccuracies can be attributed to the nonlinear part Q , while the small signal model is the linear part G . Moreover, in actual motor system, no step like changes in voltage, current, or flux are possible, so the state equations of the HFO are smooth and continuous. Therefore, it can be explained that when the eigenvalues of (24) all have negative real parts, the system is stable. Although the method based on virtual flux signal injection may cause the estimated angle to converge to a point with a large error at low speed, the system is still stable, and this has been verified by actual experiments.

REFERENCES

- [1] G. Wang, M. Valla, and J. Solsona, "Position sensorless permanent magnet synchronous machine drives—A review," *IEEE Trans. Ind. Electron.*, vol. 67, no. 7, pp. 5830–5842, Jul. 2020.
- [2] Z. Wu, W. Hua, C. Cheng, H. Zhang, and M. Hu, "A square-wave voltage injection sensorless control for dual three-phase IPMs robust to open-circuit faults," *IEEE Trans. Ind. Electron.*, vol. 71, no. 2, pp. 1113–1124, Feb. 2024.
- [3] Z. Lin, X. Li, Z. Wang, T. Shi, and C. Xia, "Minimization of additional high-frequency torque ripple for square-wave voltage injection IPMSM sensorless drives," *IEEE Trans. Power Electron.*, vol. 35, no. 12, pp. 13345–13355, Dec. 2020.
- [4] P. L. Xu and Z. Q. Zhu, "Novel square-wave signal injection method using zero-sequence voltage for sensorless control of PMSM drives," *IEEE Trans. Ind. Electron.*, vol. 63, no. 12, pp. 7444–7454, Dec. 2016.
- [5] B. Xia et al., "An improved high-frequency voltage signal injection-based sensorless control of IPMSM drives with current observer," *IEEE Trans. Transport. Electric.*, vol. 10, no. 3, pp. 5155–5167, Sep. 2024.
- [6] X. Li et al., "Low-speed rotating restart and speed recording for free-running sensorless IPMSM based on ultrahigh frequency sinusoidal wave injection," *IEEE Trans. Power Electron.*, vol. 37, no. 12, pp. 15245–15259, Dec. 2022.
- [7] X. Qiu, Y. He, W. Zhao, M. Sun, D. Zhou, and S. Jin, "Acoustic noise reduction for low-speed sensorless operation of dual three-phase PMSM with random winding injection," *IEEE Trans. Energy Convers.*, vol. 38, no. 4, pp. 2612–2622, Dec. 2023.
- [8] G. Zhang et al., "Pseudorandom-frequency sinusoidal injection for position sensorless IPMSM drives considering sample and hold effect," *IEEE Trans. Power Electron.*, vol. 34, no. 10, pp. 9929–9941, Oct. 2019.
- [9] G. Zhang, S. Li, B. Li, Q. Wang, G. Wang, and D. Xu, "Acoustic noise reduction using half-period-switching pseudorandom sinusoidal injection with external tangent demodulation for sensorless PMSM drives," *IEEE Trans. Transport. Electric.*, vol. 11, no. 2, pp. 5312–5323, Apr. 2025.
- [10] A. Benevieri, A. Formentini, M. Marchesoni, M. Passalacqua, and L. Vaccaro, "Sensorless control with switching frequency square wave voltage injection for SPMSM with low rotor magnetic anisotropy," *IEEE Trans. Power Electron.*, vol. 38, no. 8, pp. 10060–10072, Aug. 2023.
- [11] Z. Zhang, S. Zhou, H. Li, and J. Qiu, "A novel demodulation for pulsating carrier voltage injection-based encoderless control," *IEEE Trans. Power Electron.*, vol. 40, no. 1, pp. 118–123, Jan. 2025.
- [12] R. Raute, C. Caruana, C. S. Staines, J. Cilia, M. Sumner, and G. M. Asher, "Sensorless control of induction machines at low and zero speed by using PWM harmonics for rotor-bar slotting detection," *IEEE Trans. Ind. Appl.*, vol. 46, no. 5, pp. 1989–1998, Sep./Oct. 2010.
- [13] Z. Wang, W. Sun, D. Jiang, and R. Qu, "Observability analysis of Q-MRAS drive of induction motor with and without virtual voltage injection," in *2019 22nd Int. Conf. Elect. Machines Syst. (ICEMS)*, Harbin, China, 2019, pp. 1–6.
- [14] I. Boldea, M. C. Paicu, and G.-D. Andreescu, "Active flux concept for motion-sensorless unified AC drives," *IEEE Trans. Power Electron.*, vol. 23, no. 5, pp. 2612–2618, Sep. 2008.
- [15] K. Wang, W. Yao, K. Lee, and Z. Lu, "Regenerating mode stability improvements for combined voltage and current mode flux observer in speed sensorless induction machine control," *IEEE Trans. Ind. Appl.*, vol. 50, no. 4, pp. 2564–2573, Jul./Aug. 2014.
- [16] Y. Lee and S.-K. Sul, "Model-based sensorless control of an IPMSM with enhanced robustness against load disturbances based on position and speed estimator using a speed error," *IEEE Trans. Ind. Appl.*, vol. 54, no. 2, pp. 1448–1459, Mar./Apr. 2018.
- [17] V. Repecho, J. B. Waqar, D. Biel, and A. Dòria-Cerezo, "Zero speed sensorless scheme for permanent magnet synchronous machine under decoupled sliding-mode control," *IEEE Trans. Ind. Electron.*, vol. 69, no. 2, pp. 1288–1297, Feb. 2022.
- [18] J. Lee, J. Hong, K. Nam, R. Ortega, L. Praly, and A. Astolfi, "Sensorless control of surface-mount permanent-magnet synchronous motors based on a nonlinear observer," *IEEE Trans. Power Electron.*, vol. 25, no. 2, pp. 290–297, Feb. 2010.
- [19] Y. Pang, J. M. Knežević, D. Glose, and C. M. Hackl, "Sensorless control of electrically excited synchronous machines considering magnetic saturation and flux linkage dynamics for automotive applications," *IEEE Trans. Ind. Electron.*, vol. 71, no. 1, pp. 104–114, Jan. 2024.
- [20] Q. Lin, L. Liu, and D. Liang, "Hybrid active flux observer to suppress position estimation error for sensorless IPMSM drives," *IEEE Trans. Power Electron.*, vol. 38, no. 1, pp. 872–886, Jan. 2023.
- [21] P. Wang, Z. Q. Zhu, and D. Liang, "Virtual signal injection-based online full-parameter estimation of surface-mounted PMSMs without influence of position error and inverter nonlinearity," *IEEE Trans. Emerg. Sel. Topics Power Electron.*, vol. 13, no. 2, pp. 2341–2357, Apr. 2025.
- [22] W. Sun, Z. Wang, D. Xu, and B. Wang, "Robust stability improvement for speed sensorless induction motor drive at low speed range by virtual voltage injection," *IEEE Trans. Ind. Electron.*, vol. 67, no. 4, pp. 2642–2654, Apr. 2020.
- [23] P. Koothu Kesavan, U. Subramaniam, and D. J. Almakhlis, "Reduced switch inverter-fed sensorless PMSM drive with multistage MRAS speed estimator based on stator flux and electromagnetic torque," *IEEE Trans. Transport. Electric.*, vol. 10, no. 2, pp. 2620–2631, Jun. 2024.
- [24] P. Vaclavek, P. Blahak, and I. Herman, "AC drive observability analysis," *IEEE Trans. Ind. Electron.*, vol. 60, no. 8, pp. 3047–3059, Aug. 2013.
- [25] M. Koteich, A. Maloum, G. Duc, and G. Sandou, "Discussion on 'AC drive observability analysis'," *IEEE Trans. Ind. Electron.*, vol. 62, no. 11, pp. 7224–7225, Nov. 2015.
- [26] M. Hinkkanen, T. Tuovinen, L. Harnefors, and J. Luomi, "A combined position and stator-resistance observer for salient PMSM drives: Design and stability analysis," *IEEE Trans. Power Electron.*, vol. 27, no. 2, pp. 601–609, Feb. 2012.
- [27] O. Wallmark and L. Harnefors, "Sensorless control of salient PMSM drives in the transition region," *IEEE Trans. Ind. Electron.*, vol. 53, no. 4, pp. 1179–1187, Jun. 2006.
- [28] A. Piippo, M. Hinkkanen, and J. Luomi, "Analysis of an adaptive observer for sensorless control of interior permanent magnet synchronous motors," *IEEE Trans. Ind. Electron.*, vol. 55, no. 2, pp. 570–576, Feb. 2008.
- [29] S. Wang et al., "Flux observer based on enhanced second-order generalized integrator with limit cycle oscillator for sensorless PMSM drives," *IEEE Trans. Power Electron.*, vol. 38, no. 12, pp. 15982–15995, Dec. 2023.
- [30] R. Hermann and A. Krener, "Nonlinear controllability and observability," *IEEE Trans. Autom. Control*, vol. 22, no. 5, pp. 728–740, Oct. 1977.

- [31] S. Ichikawa, M. Tomita, S. Doki, and S. Okuma, "Sensorless control of permanent-magnet synchronous motors using online parameter identification based on system identification theory," *IEEE Trans. Ind. Electron.*, vol. 53, no. 2, pp. 363–372, Apr. 2006.
- [32] S. Liu, W. Sun, and R. Qu, "Adaptive compensation method for D-axis current reference for speed sensorless induction motor drives with virtual voltage injection," in *2024 IEEE 7th Int. Elect. Energy Conf. (CIEEC)*, Harbin, China, 2024, pp. 3546–3552.
- [33] F. Paganini, *A Course in Robust Control Theory*. Cham, Switzerland: Springer, 2014.
- [34] P. Holmes and J. Guckenheimer, *Nonlinear Oscillations Dynamical Systems and Bifurcations of Vector Fields*. Cham, Switzerland: Springer, 1985.



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